Frl 2.3

Växelström $\quad v(t) = V_m \cos(\omega t + \varphi)$

$\bar{V} = a + jb = |\bar{V}| e^{j \arg\{\bar{V}\}} = V_m e^{j(\omega t + \varphi)} = V_m e^{j\varphi} e^{j\omega t}$

Notera: $\quad |e^{jx}| = \sqrt{\cos^2(x) + \sin^2(x)} = 1$

$|V_m e^{j\varphi}| = |V_m| |e^{j\varphi}| = V_m \cdot 1$

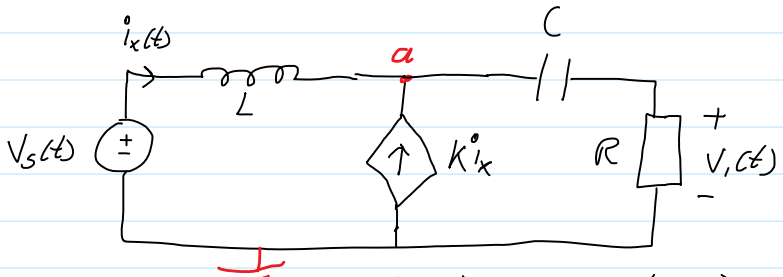
$\arg\{V_m e^{j\varphi}\} = \arg\{V_m\} + \arg\{e^{j\varphi}\} = 0 + \varphi$

$\arg\left\{\frac{\bar{z}_1}{\bar{z}_2}\right\} = \arg\{\bar{z}_1\} - \arg\{\bar{z}_2\}$

Idag - Växelström (Nodanalys, superpos.)

Antar linjära kretsar, stationärt tillstånd,
$\Rightarrow$ jω-metoden

Exempel

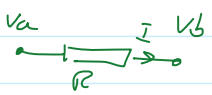


Bestäm $V_1(t)$ $\quad V_s(t) = V_0 \cos(\omega t + \alpha)$

$(x(t) \longrightarrow \bar{X} \Rightarrow \ldots \bar{Y} \longrightarrow y(t)$ - arbetsgång)

KCL:a $\quad \frac{\bar{V}_a - \bar{V}_s}{j\omega L} + \frac{\bar{V}_a - 0}{R + \frac{1}{j\omega C}} - k\bar{I}_x = 0$

$\bar{Z} = R + \frac{1}{j\omega C}$ $\quad$ & $\quad \bar{I}_x = \frac{\bar{V}_s - \bar{V}_a}{j\omega L}$



$\bar{I} = \frac{V_a - V_b}{R}$

$\frac{\bar{V}_a - \bar{V}_s}{j\omega L} + \frac{\bar{V}_a}{\bar{Z}} - k\left(\frac{\bar{V}_a - \bar{V}_s}{j\omega L}\right) = 0$

$\bar{V}_a \left(\frac{(1+k)\bar{Z} + j\omega L}{j\omega L \bar{Z}}\right) = \bar{V}_s \left(\frac{1+k}{j\omega L}\right) \Rightarrow$

V_a \ $j\omega L$ ... $j\omega L$ ...

$$\bar{V}_a = \frac{\bar{V}_S}{1 + \frac{j\omega L}{(1+k)(R + \frac{1}{j\omega C})}}$$

Vi söker $V_1(t)$ → behöver $\bar{V}_1$

$$\bar{V}_1 = R\bar{I}_{ab} = R\frac{\bar{V}_a - 0}{R + \frac{1}{j\omega C}} \Rightarrow$$

$$\bar{V}_1 = \frac{R}{R + \frac{1}{j\omega C}}\left(\frac{1}{1 + \frac{j\omega L}{(1+k)(R + \frac{1}{j\omega C})}}\right)\bar{V}_S$$

$\left(\bar{V}_1 = \frac{a_1 + jb_1}{a_2 + jb_2}\right)$

$V_S(t) = V_0\cos(\omega t + \alpha)$ cosinus som "riktfas" $\Rightarrow$

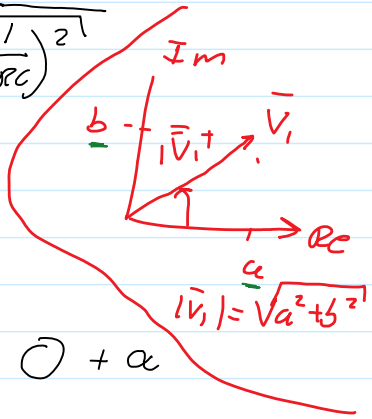
$$\bar{V}_S = V_0 e^{j\alpha} \Longrightarrow$$

$$\bar{V}_1 = \frac{V_0 e^{j\alpha}}{1 + j\left(\frac{\omega L}{(1+k)R} - \frac{1}{\omega RC}\right)}$$

(Tips $\bar{Z} = (\cdots) + j(\cdots) \rightarrow |\bar{Z}|$, $\arg\{\bar{Z}\}$)

$V_1(t)$ söks, $|\bar{V}_1|$ & $\arg\{\bar{V}_1\}$ behövs → skriv $\bar{V}_1$ på polärform

$$|\bar{V}_1| = \left|\frac{V_0 e^{j\alpha}}{1 + j(\cdots)}\right| = \frac{V_0 \cdot 1}{\sqrt{1^2 + \left(\frac{\omega L}{(1+k)R} - \frac{1}{\omega RC}\right)^2}}$$



$$\arg\{\bar{V}_1\} = \arg\left\{\frac{V_0 e^{j\alpha}}{1 + j(\cdots)}\right\}$$

$$\arg\{V_0 e^{j\alpha}\} = \arg\{V_0\} + \arg\{e^{j\alpha}\} = 0 + \alpha$$

[antar:

$$\arg\{V_0 e^{j\alpha}\} = \arg\{V_0\} + \arg\{e^{j\alpha}\} = 0 + \alpha$$

$$\arg\{1 + j(\ldots)\} = \left[\text{antar } \frac{\omega L}{(1+k)R} - \frac{1}{\omega RC} > 0\right] = \tan^{-1}\left(\frac{Im}{Re}\right) =$$

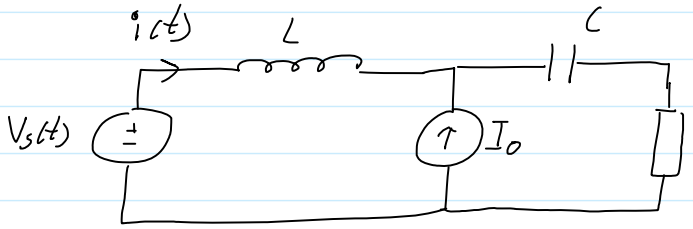
$$= \tan^{-1}\left(\frac{\omega L}{(1+k)R} - \frac{1}{\omega RC}\right) \quad \Rightarrow \quad \arg\{\bar{V}_1\} =$$

$$\alpha - \tan^{-1}(\ldots)$$

$$\bar{V}_1 = |\bar{V}_1| e^{j\arg\{\bar{V}_1\}} \quad \Rightarrow$$

$$v_1(t) = \frac{V_0}{\sqrt{1 + \left(\frac{\omega L}{(1+k)R} - \frac{1}{\omega RC}\right)^2}} \cos\left(\omega t + \alpha - \tan^{-1}\left(\frac{\omega L}{(1+k)R} - \frac{1}{\omega RC}\right)\right)$$

Exempel Superposition



$V_s(t) = V_0 \cos(\omega t + \alpha)$

I_0 - DC källa

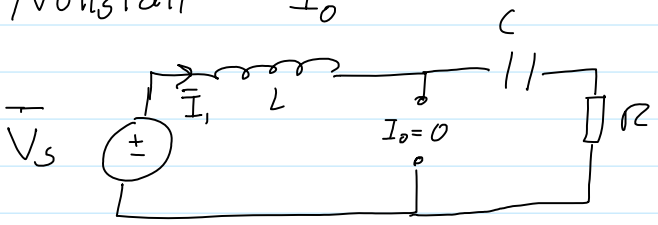
Beräkna $i(t)$

Eftersom källor med olika frekvenser → superpos

$$i(t) = i_1(t) + i_2(t)$$

$V_s(t)$ I_0

Nollställ I_0



$\left(\frac{1}{j} = -j\right)$

KVL: $+\bar{V}_s - j\omega L\bar{I}_1 - \frac{1}{j\omega C}\bar{I}_1 - R\bar{I}_1 = 0 \iff$

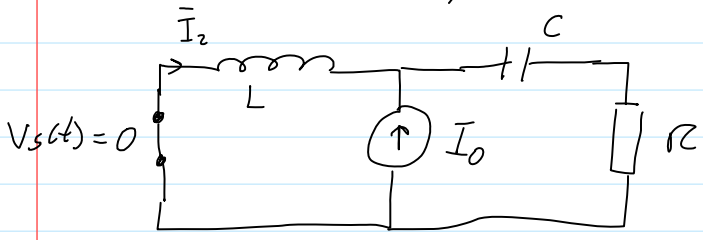
$$\bar{I}_1 = \frac{V_0 e^{j\alpha}}{R + j\left(\omega L - \frac{1}{\omega C}\right)} \quad \Rightarrow \left(\omega L - \frac{1}{\omega C} > 0\right)$$

$$i_1(t) = |\bar{I}_1| \cos\left(\omega t + \arg\{\bar{I}_1\}\right) =$$

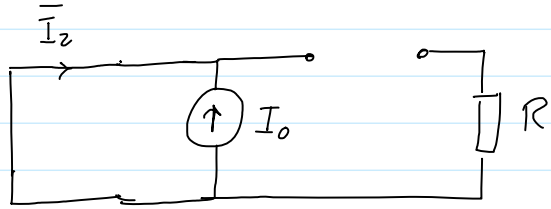
$$i_1(t) = |\underline{I}_1| \cos(\omega t + \arg\{\underline{I}_1\}) =$$

$$= \frac{V_0}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \cos\left(\omega t + \left(\alpha - \tan^{-1}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)\right)\right)$$

Söker nu $i_2(t)$ Nollställ $V_s(t)$



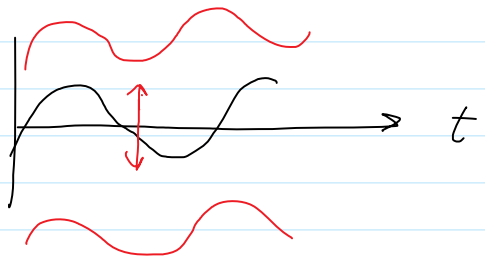
I_0 - DC källa ! =>



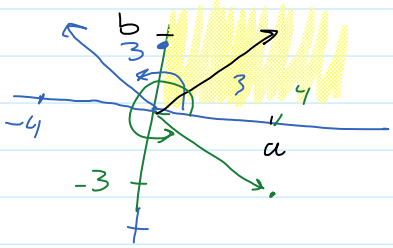
$$\Rightarrow i_2(t) = -I_0$$

$$i(t) = i_1(t) + i_2(t) =$$

$$= \frac{V_0}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \cos\left(\omega t + \alpha - \tan^{-1}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)\right) - I_0$$



$\tan^{-1}\left(\frac{-3}{4}\right)$



$b > 0$
$a > 0$

$\tan^{-1}\left(\frac{3}{-4}\right)$

$\tan^{-1}\{Im/Re\}$ ← $\bar{V} = 3\angle -120°$